HridayTwin

A personal cardiometabolic digital twin for Indian adults with diabetes and hypertension

Team Winmasters
Manjit Kalita · Dhruvik Sorathiya · Bishwarup Roy
Goa Institute of Management

Happiest Health · Digital Twin Challenge 2026 · Proof of concept

Research prototype using synthetic patients only. Not a medical device; not for clinical use.

1. The problem

Cardiometabolic care has a visibility gap between clinic visits.

- Adults with diabetes and/or hypertension may generate daily signals, but clinicians often see them only at infrequent appointments.
- Medication adherence, meals, fasting, illness, activity and home blood pressure can change risk between visits.
- A useful support system should identify who may need follow-up, explain contributing signals, and let clinicians explore candidate plans safely.

Problem framing: improve between-visit visibility and care-team prioritisation—not replace clinician judgment.

2. The healthcare use case

HridayTwin focuses on simulated Indian outpatient archetypes.

• Inputs: electronic health record context, medication list, wearable activity/heart-rate data, pill-box adherence, CGM/glucometer readings, home BP, diet and calendar context.

• Use cases: review near-term event risk, understand personal response trends, inspect alerts, and compare what-if plans.

• The dashboard demonstrates eight held-out synthetic patient archetypes and a 180-day replay.

The current prototype is a research demonstration. It does not connect to real patients, hospitals, or live medical devices.

3. Proposed solution

One workflow combines a personal twin, risk engine and clinician console.

- Personal twin: Bayesian linear state-space models estimate glucose and systolic blood pressure trends and update daily with a Kalman filter.
- Risk engine: gradient-boosting model estimates 14-day event risk and event-type risks; SHAP-style explanations highlight drivers.
- What-if engine: Monte Carlo roll-outs compare candidate care plans over 90 days, with safety guardrails and clinician-review labels.

Design rule: the twin simulates; the risk engine judges. Twin-state features were excluded after an ablation worsened AUROC.

4. System architecture

The project uses a modular, reproducible pipeline.

- Synthetic data generator creates EHR context, daily streams, latent physiology and event labels.
- Feature pipeline creates causal windows; training and validation use patient-level splits and leakage checks.
- Exports include the static clinician dashboard, metrics, model card, data dictionary, validation plan and sample FHIR bundle.

> See the companion file HridayTwin_Architecture_Diagram.pdf and docs/ARCHITECTURE.md.

5. Technical stack

- Python 3.10+; NumPy and pandas for synthetic data and features.
- Custom Bayesian linear state-space twin with daily Kalman updates.
- scikit-learn HistGradientBoostingClassifier and logistic-regression baseline.
- SHAP TreeExplainer for model explanations; HL7 FHIR R4-oriented interoperability artifacts.
- Vanilla HTML, CSS and JavaScript dashboard with hand-built SVG charts; pytest and GitHub Actions for quality checks.

6. Model validation and reported results

Results are from the documented synthetic cohort; they are not clinical evidence.

Measure	Reported result
Synthetic cohort	1,200 patients × 180 days
Fused vs EHR-only AUROC	0.726 vs 0.708; +0.017, 95% CI +0.004 to +0.032; patient-level bootstrap p = 0.008
Threshold rules baseline	AUROC 0.615; fused model +0.111
Severe hypoglycaemia	AUROC 0.943 fused vs 0.907 EHR-only
Alert operating point	10 alerts per 100 patient-weeks; 2.9× lift over base rate; median warning 7 days
Leakage control	Label-permutation AUROC 0.515, close to chance

Interpret cautiously: the cohort is synthetic. Real-world retrospective and prospective validation is still required.

7. Dashboard walkthrough

Suggested sequence for the live console demo.

- Open the public dashboard and point out the synthetic-data safety banner.
- Select a patient archetype; explain the displayed risk level and context.
- Start replay and show how risk, alerts and twin expectation bands change over time.
- Compare observed glucose/BP with the twin's expectation; inspect the alert feed and risk drivers.
- Use the what-if planner to compare candidate plans and read the guardrails and clinician-review warning.

Public dashboard: https://dhruvikms.github.io/hridaytwin/dashboard/

8. Limitations, safety and ethics

- No real patient data are used; the dashboard is not a medical device and must not be used for clinical decisions.
- Twin-state features reduced model performance in ablation; individual step sensitivity was not learned.
- Acute cardiovascular-event performance did not improve with the stream; timing the dangerous week remains only modestly better than chance.
- Before deployment: independent validation, ethics approval, silent-mode pilot, clinical oversight, safety monitoring and regulatory review.

Every recommendation is for clinician review. No autonomous treatment action is supported.

9. Roadmap and conclusion

- Stage 1: retrospective validation on paired EHR and device data from an Indian partner under ethics approval.
- Stage 2: prospective silent-mode pilot; compare predictions with outcomes without showing recommendations to clinicians.
- Stage 3: clinician-in-the-loop pilot with alert-fatigue, bias, safety and workflow monitoring.
- HridayTwin demonstrates an end-to-end research architecture for turning longitudinal synthetic health signals into interpretable review workflows.

Team Winmasters · Goa Institute of Management